S.C.O.U.T. · ELECTRICAL · DRAWING SET

Electronics system: block diagram, stack-up, wiring and takeoffs

Everything that goes inside and around the buoy's electronics housing, how each part connects to the Feather M0, which cable or connector each link uses, and what is still to buy. Prototype design as of 2026-10-12. Values marked † are proposed and get confirmed on the bench.

PRELIMINARY · NOT FOR FINAL BUILD

PROJECT	**S.C.O.U.T. nearshore reef buoy**
DRAWING NO.	SCOUT-E-100
REVISION	B (draft) — audio switch to A3, 6 m cable
DATE	2026-10-12
SCALE	NTS
DRAWN	I. Rodriguez (ECEN)
CHECKED	— (S. Krishnan to review)
SHEETS	0 Hydrophone · 1 Block diagram · 2 Stack-up · 3 Pins & cables · 4 Material takeoff · 5 Equipment takeoff · 6 Notes & references · 7 Every connection

Sheet 0 — Hydrophone selection: Aquarian H2dM

Why this one

✓ **Fits the 3.3 V system:** it runs on plug-in power, at least 2 V and 0.7 mA. 3.3 V through 1.1 kΩ gives about 2.5 V. No 5 V or 48 V rail needed.

✓ **Easy to wire:** it has a standard 3.5 mm TRS plug. It goes into a jack breakout on the Proto board, so there is nothing to solder on the hydrophone itself.

✓ **Easy to read:** the output is a plain analog voltage (1.1 kΩ output impedance). A two-stage MCP6002 preamp feeds the M0's ADC on A2. No codec chip and no I²S.

✓ **Good enough for reef sound:** it covers below 10 Hz to above 100 kHz, which includes fish calls (50 Hz–3 kHz) and snapping shrimp (2–20 kHz) up to our 11 kHz sampling limit.

✓ **Rated well past our depth:** under 80 m, against a reef depth of about 1–5 m. It weighs 105 g and measures 25 × 46 mm.

✓ **Good price:** $229 base, the lowest-cost calibrated hydrophone we found that fits. The cable is 3, 6, 9 or 15 m; **we are ordering 6 m** (covers 3–6 m water depth).

Datasheet: H2dM user's guide, saved as `DATASHEETS/aquarian-h2dm-manual.pdf`. Order from aquarianaudio.com and confirm the cable-option price at checkout.

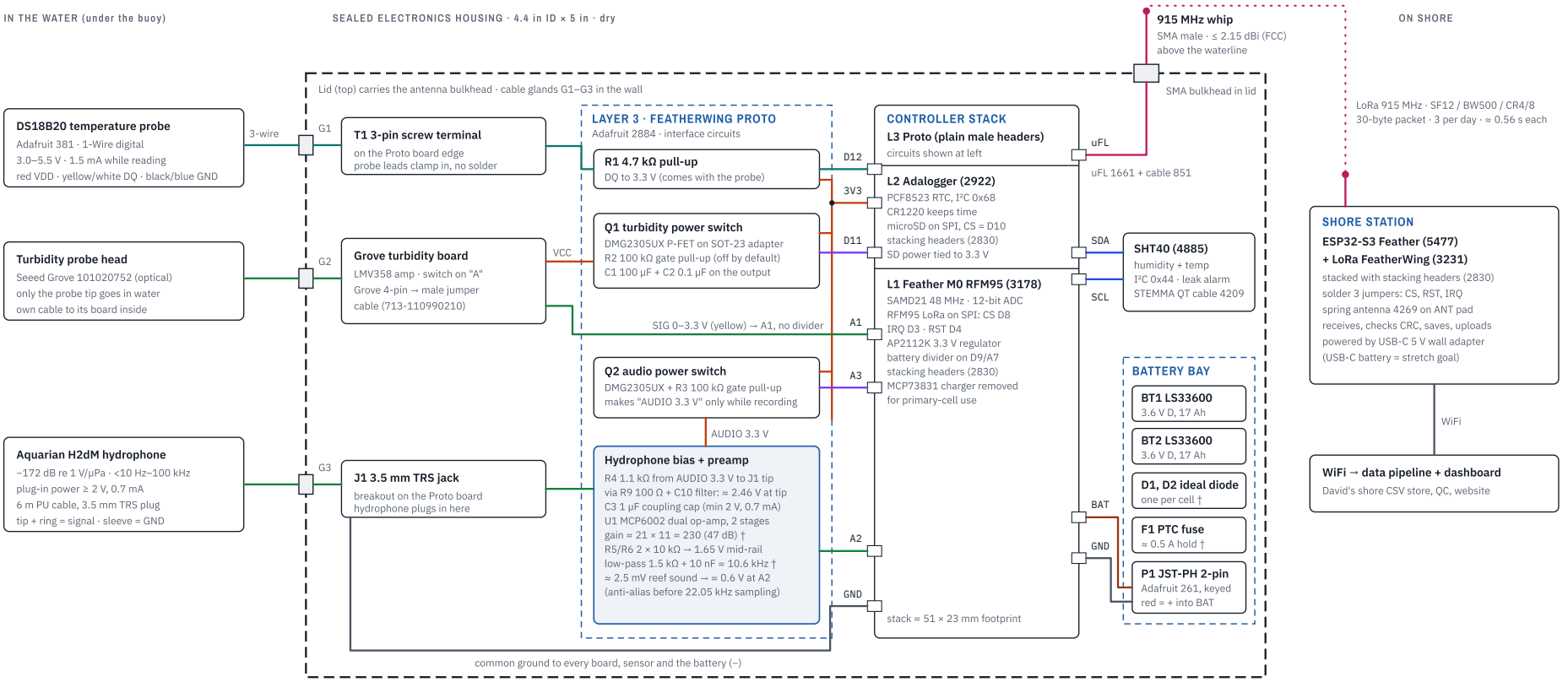
Options compared

OPTION	POWER	OUTPUT	VERDICT
Aquarian H2dM	Plug-in, ≥ 2 V, 0.7 mA	3.5 mm analog	SELECTED
Aquarian H2dX (XLR)	Phantom power, up to 48 V	XLR balanced	Rejected: needs a high-voltage supply
Aquarian H2a	Plug-in	3.5 mm	Discontinued (replaced by the H2d)
HydroMoth	Own battery	Records to its own SD card	Not integrated with the M0; can't schedule or summarize (unverified)
Aquarian A5 element	—	Bare element	Needs our own potting and preamp (unverified, about €199)
DIY piezo disc	—	Uncalibrated	Too cheap: no sensitivity spec, so it gives no scientific numbers

Signal math: −172 dB re 1 V/µPa at a 120 dB re 1 µPa reef gives 10^((120−172)/20) ≈ 2.5 mV. Gain of about 230 (47 dB) brings that to about 0.6 V, centered on 1.65 V, which leaves headroom for loud events †.

Sheet 1 — System block diagram

Battery 3.6 V · 3.3 V rail · Ground · 1-Wire data · Analog signal · I²C · GPIO on/off control · RF / LoRa



Inside the stack (no wires to add): SPI (SCK/MOSI/MISO) is shared by the LoRa radio (CS D8) and the SD card (CS D10); I²C (SDA/SCL) is shared by the clock (0x68) and the SHT40 (0x44).

Firmware wakes every 30 min on the SAMD21 internal RTC (standby), switches sensors on with D11/A3 LOW, reads, logs to SD, transmits in fixed slots, switches everything off.

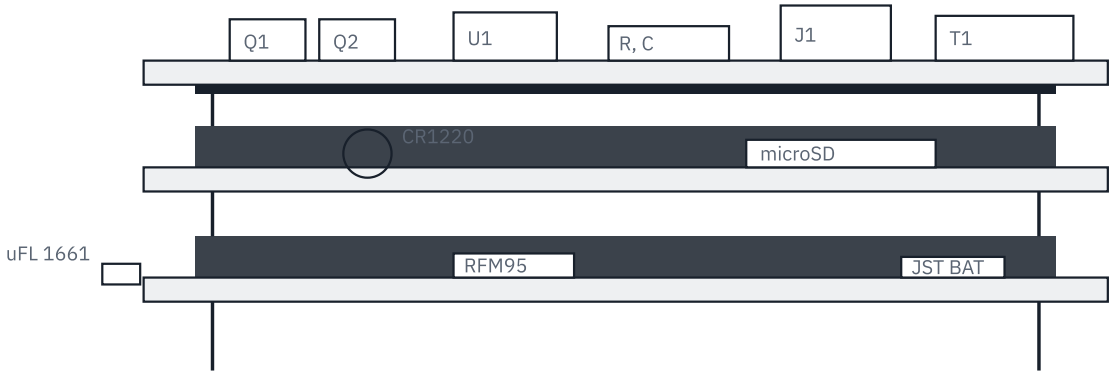
Pins already used by the board: D3, D4, D8 (radio), D9/A7 (battery), D10 (SD), D13 (red LED, keep off). D5 = PCF8523 INT (firmware). Free spares: A0, A4, A5, D6 (keep free for radio IO1); D0/D1 are Serial1.

† proposed value; confirm on the bench before soldering. G1–G3: cable glands from lab stock; the 3.5 mm plug (≈ 6 mm) must pass G3 or be fitted after the gland.

Sheet 2 — Controller stack-up and battery pack

Three Feather-format boards stack on headers: the Feather M0 at the bottom, the Adalogger in the middle, the FeatherWing Proto on top. Stacking headers (female socket on top, long pins below) let each board pass every pin up to the next. The Proto board takes plain male pins because nothing stacks on it.

SIDE VIEW (not to scale)



L3 · FeatherWing Proto (2884)
plain male headers down into L2 · Q1/Q2 on SOT-23 adapters, U1 MCP6002, J

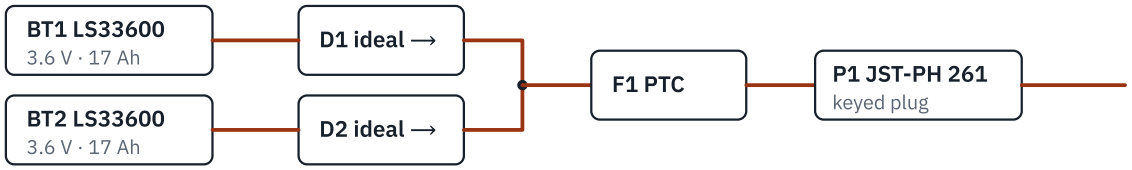
L2 · Adalogger FeatherWing (2922)
stacking headers (2830): sockets on top for L3, long pins down into L1

L1 · Feather M0 RFM95 (3178)
stacking headers (2830): long pins below go into a breadboard on the bench, s

Antenna: bench = spring 4269 on the ANT pad; deployed = uFL 1661 → cable 851 → SMA bulkhead → whip. Use one, never both.

Approximate stack height ≈ 30 mm with two stacking-header levels †. Footprint 51 × 23 mm (Feather standard).

BATTERY PACK (schematic)



→ Feather BAT
cells in PARALLEL (never series: 7.2 V is too high)
each cell's ideal diode stops one cell charging the other
the fuse limits a short; the keyed plug can't go in backwards

Sheet 3 — Pin assignment and cable schedule

Feather M0 pin assignment

PIN	USED BY	TYPE	SOURCE OF TRUTH
D3 · D4 · D8	RFM95 radio IRQ · RST · CS	SPI + GPIO	Feather M0 schematic
SCK · MOSI · MISO	Radio + SD card (shared)	SPI	Schematic; Adalogger guide p. 7
D10	SD card chip select	SPI CS	Adalogger guide p. 7
SDA · SCL	PCF8523 (0x68) + SHT40 (0x44)	I²C	Adalogger p. 6; SHT40 guide
D9 / A7	Battery divider (100 k / 100 k)	Analog	Schematic R3/R6
D12	DS18B20 data (DQ)	1-Wire	Firmware config.h (decided 2026-10-12; update schematic)
A1	Turbidity signal	Analog	SCO-85
D11	Q1 gate (turbidity power, LOW = on)	GPIO	This drawing
A3	Q2 gate (audio power, LOW = on)	GPIO	This drawing (A3 used as digital out; D5 is taken by RTC INT)
A2	Hydrophone preamp output	Analog	This drawing
D5	PCF8523 INT1 wake (backup to internal RTC)	GPIO in	Firmware config.h
D13	Red LED (keep off)	GPIO	Schematic
BAT · GND	Battery pack via JST-PH	Power	Schematic

Cable and connection schedule

ID	FROM → TO	CABLE / CONNECTOR	HOW TO CONNECT
W1	DS18B20 → T1	Probe's own 3-core lead through G1	Strip, lightly tin, clamp in terminal (red 3V, yellow/white D12, black/blue GND); R1 4.7 kΩ DQ→3.3 V on Proto
W2	Turbidity probe → Grove board	Probe's own cable through G2	Plugs into the Grove board; seal the probe top
W3	Grove board → Proto / M0	Grove → male jumper 110990210	Red→Q1 out, black→GND, yellow→A1, white unused
W4	H2dM → J1	6 m PU cable, 3.5 mm TRS through G3	Plug into J1; tip and ring both carry signal, sleeve GND
W5	SHT40 → stack	STEMMA QT → male 4209	Red 3.3 V, black GND, blue SDA, yellow SCL
W6	Battery pack → BAT	JST-PH 2-pin 261	Red +, black –; plug only after the charger is removed
W7	M0 uFL → lid	uFL 1661 + SMA cable 851	Solder 1661 on the M0 uFL pads; 851 bulkhead nut in the lid
W8	Bulkhead → whip	SMA male whip, ≤ 2.15 dBi	Hand-tighten; never transmit with nothing attached
W9	Shore station power	USB-C data cable + 5 V adapter	Wall outlet at the shore site

Sheet 4 — Material takeoff

Every part by section. Prices are list prices per unit; ranges are estimates for parts not yet quoted. The Adafruit order (invoice 3758294) has been placed; its shipping and tax are listed once at the end.

ITEM	PART NUMBER	DESCRIPTION	QTY	UNIT	EXTENDED	STATUS	SOURCE
1 · Controller stack (buoy)							
M0	Adafruit 3178	Feather M0 RFM95 LoRa 900 MHz	1	$34.95	$34.95	ORDERED	Adafruit
AD	Adafruit 2922	Adalogger FeatherWing (RTC + microSD)	1	$8.95	$8.95	ORDERED	Adafruit
PR	Adafruit 2884	FeatherWing Proto (interface circuits)	1	$4.95	$4.95	ORDERED	Adafruit
SH	Adafruit 2830	Stacking headers, 12 + 16 pin (M0, Adalogger)	2	$1.25	$2.50	ORDERED	Adafruit
SD	Adafruit 5250	128 MB microSD (bench logging)	1	$3.95	$3.95	ORDERED	Adafruit
CC	Mouser 658-CR1220	CR1220 3 V coin cell (RTC backup)	1	$1.01	$1.01	MOUSER	Mouser
Subtotal — Controller stack (buoy)					**$56.31**		
2 · Sensors							
TS	Adafruit 381	DS18B20 waterproof probe (incl. 4.7 kΩ)	1	$9.95	$9.95	ORDERED	Adafruit
TB	Mouser 713-101020752	Seeed Grove turbidity sensor (3.3 V)	1	$20.80	$20.80	MOUSER	Mouser
TC	Mouser 713-110990210	Grove → male jumper cable, 5-pack	1	$3.19	$3.19	MOUSER	Mouser
HS	Adafruit 4885	SHT40 humidity + temperature	1	$5.95	$5.95	ORDERED	Adafruit
HC	Adafruit 4209	STEMMA QT → male header cable	1	$0.95	$0.95	ORDERED	Adafruit
HY	Aquarian H2dM	H2dM hydrophone, 3.5 mm PIP, 6 m cable, decided 2026-10-12 (base $229 + cable option)	1	$229.00–249.00	$229.00–249.00	LATER	Aquarian Audio
Subtotal — Sensors					**$269.84–289.84**		

ITEM	PART NUMBER	DESCRIPTION	QTY	UNIT	EXTENDED	STATUS	SOURCE
3 · Interface circuits (on the Proto board)							
Q1, Q2	Mouser 621-DMG2305UX-7	P-MOSFET, SOT-23 (sensor + audio power)	2	$0.40	$0.80	MOUSER	Mouser
AD1	SparkFun BOB-00717	SOT-23 to DIP adapter	2	$1.25	$2.50	MOUSER	SparkFun / Mouser
U1	MCP6002-I/P	Dual rail-to-rail op-amp, DIP-8 (preamp)	1	$0.40–0.60	$0.40–0.60	LATER	DigiKey / Mouser
J1	Adafruit 1699	3.5 mm stereo jack breakout (L = tip)	1	$0.95	$0.95	LATER	Adafruit
T1	3-pin 5 mm screw terminal	DS18B20 lead terminal	1	$0.50–1.00	$0.50–1.00	LAB	Lab
R1–R13	¼ W 1 % metal film	4.7 k (with probe), 100 k ×3, 10 k ×3, 1 k ×2, 1.1 k, 1.5 k, 20 k, 100 Ω	1	$0.00–10.00	$0.00–10.00	LAB	Lab or resistor kit
C1–C10	Capacitors	100 µF 10 V elec.; 10 µF ×5; 1 µF ceramic; 0.1 µF ×2; 10 nF	1	$0.00–10.00	$0.00–10.00	LAB	Lab or cap kit
U1S	DIP-8 socket	So the op-amp can be swapped	1	$0.00–0.50	$0.00–0.50	LAB	Lab
H3	Male header 12 + 16 pin	Proto → Adalogger (if the Proto ships without)	1	$0.00–0.50	$0.00–0.50	LAB	Lab / Adafruit
WR1	22 AWG solid hookup wire	Point-to-point wiring on the Proto board	1	$0.00–16.00	$0.00–16.00	LAB	Lab / Adafruit
WR2	22 AWG stranded, red + black	Battery pack leads	1	$0.00–8.00	$0.00–8.00	LAB	Lab
Subtotal — Interface circuits (on the Proto board)					**$5.15–50.85**		
4 · Power							
BT	Saft LS33600 (solder tabs)	3.6 V Li-$SOCl_2$ D cell, 17 Ah	2	$22.00–35.00	$44.00–70.00	LATER	BatteryGuy / DigiKey
D1, D2	TI LM66100 (SC-70-6)	Ideal diode per cell	2	$0.60–1.20	$1.20–2.40	LATER	DigiKey / Mouser

ITEM	PART NUMBER	DESCRIPTION	QTY	UNIT	EXTENDED	STATUS	SOURCE
AD2	SC-70-6 to DIP adapter	Lets the LM66100 go on perfboard	2	$1.50–4.00	$3.00–8.00	LATER	DigiKey / Chip Quik
F1	Bourns MF-R050 PTC (0.5 A hold) †	Resettable fuse at the pack	1	$0.50–1.50	$0.50–1.50	LATER	DigiKey / Mouser
P1	Adafruit 261	JST-PH 2-pin cable (pack plug + bench lead)	2	$0.75	$1.50	ORDERED	Adafruit
BB	Adafruit 4654	MiniBoost 5 V (backup only)	1	$3.95	$3.95	ORDERED	Adafruit
HZ	Hazmat shipping	Lithium-metal cells ship ground only	1	$10.00–30.00	$10.00–30.00	LATER	Seller
Subtotal — Power					**$64.15–117.35**		
5 · RF / antenna							
A1	Adafruit 4269	Spring antenna 915 MHz (buoy + shore bench)	2	$0.95	$1.90	ORDERED	Adafruit
A2	Adafruit 1661	uFL SMT connector (on the M0)	1	$0.75	$0.75	ORDERED	Adafruit
A3	Adafruit 851	SMA (panel) to uFL cable	1	$3.95	$3.95	ORDERED	Adafruit
A4	915 MHz whip, SMA male, ≤ 2.15 dBi	Deployed antenna on the lid	1	$10.00–15.00	$10.00–15.00	LATER	DigiKey / Mouser
Subtotal — RF / antenna					**$16.60–21.60**		
6 · Shore station							
SS1	Adafruit 5477	ESP32-S3 Feather 4 MB / 2 MB PSRAM	1	$17.50	$17.50	ORDERED	Adafruit
SS2	Adafruit 3231	LoRa Radio FeatherWing RFM95W 900 MHz	1	$19.95	$19.95	ORDERED	Adafruit
SS3	Adafruit 2830	Stacking headers (shore stack)	1	$1.25	$1.25	ORDERED	Adafruit
SS4	USB-C data cable + 5 V adapter	Shore power	1	$0.00–10.00	$0.00–10.00	LAB	Team / lab
Subtotal — Shore station					**$38.70–48.70**		
7 · Housing electrical consumables							

ITEM	PART NUMBER	DESCRIPTION	QTY	UNIT	EXTENDED	STATUS	SOURCE
CG	Cable glands G1–G3	From lab stock (decision 2026-09-24)	3	$0.00	$0.00	LAB	Lab
MS	industrial microSD 16–32 GB	Deployment logging incl. audio	1	$12.00–18.00	$12.00–18.00	LATER	DigiKey / Mouser
CO	Silicone conformal coating	Protects boards from condensation	1	$15.00–22.00	$15.00–22.00	LATER	DigiKey / Amazon
DS	Silica-gel desiccant packs	Keeps the housing dry	1	$5.00–8.00	$5.00–8.00	LATER	Amazon
Subtotal — Housing electrical consumables					**$32.00–48.00**		
8 · Optional (recommended)							
OP1	Adafruit 3178	Second Feather M0 so firmware and hardware work in parallel	1	$34.95	$34.95	OPTIONAL	Adafruit
OP2	Spares kit	DS18B20, turbidity sensor, microSD, antenna	1	$60.00–70.00	$60.00–70.00	OPTIONAL	Adafruit / Mouser
Subtotal — Optional (recommended)					**$94.95–104.95**		
9 · Shipping and tax already paid							
SH1	Invoice 3758294	Adafruit shipping $7.70 + tax $11.98	1	$19.68	$19.68	ORDERED	Adafruit
Subtotal — Shipping and tax already paid					**$19.68**		
Spent so far (Adafruit, incl. shipping + tax)					**$142.58**		
To place now (Mouser)					**$28.30**		
Planned follow-up orders					**$331.05–425.45**		
TOTAL — prototype electronics (required)					**$502.43–652.33**		
TOTAL — including optional items					**$597.38–757.28**		

Status: ORDERED on invoice 3758294 · MOUSER order to place now · LATER planned follow-up order · LAB from lab stock · OPTIONAL not required for the prototype.

Sheet 5 — Equipment takeoff

Tools and test equipment needed to build and verify the electronics. Everything is borrowed from the lab (rooms 4017–4019, checked out through Dr. Krishnan) or is the team's own, so **no equipment purchase is planned**. The "cost if not in lab" column is only a fallback estimate in case an item turns out to be missing; "—" means lab-only equipment we would not buy.

ITEM	WHAT IT'S FOR	QTY	COST IF NOT IN LAB	SOURCE
A · Soldering and assembly				
Temperature-controlled soldering station	Headers, SOT-23 adapters, uFL connector	1	—	Lab (borrow)
Fine conical / 1 mm chisel tips	SOT-23 MOSFETs and the uFL pads	1	$0.00–8.00	Lab (borrow)
Rosin-core solder, 0.5–0.8 mm	All joints	1	$8.00–12.00	Lab (borrow)
Flux pen	Small SMD joints (SOT-23, uFL)	1	$6.00–9.00	Lab (borrow)
Desoldering braid	Removing the MCP73831 charger, fixing bridges	1	$4.00–6.00	Lab (borrow)
ESD-safe fine tweezers	Placing SOT-23 parts	1	$5.00–8.00	Lab (borrow)
Helping hands / PCB holder	Holding boards while soldering	1	$0.00–15.00	Lab (borrow)
Flush cutters + wire strippers	Leads, jumper wires	1	$0.00–18.00	Lab (borrow)
Heat-shrink assortment	Insulating battery and sensor leads	1	$6.00–9.00	Lab (borrow)
Hot-air / heat gun	Shrinking heat-shrink, removing the charger chip	1	—	Lab (borrow)
Fume extractor	Soldering safety	1	—	Lab (borrow)
Subtotal — Soldering and assembly			**$0 · fallback $29.00–85.00**	

ITEM	WHAT IT'S FOR	QTY	COST IF NOT IN LAB	SOURCE
B · Test and measurement				
Digital multimeter with µA range	Sleep current, voltages, continuity	1	–	Lab (borrow)
Current-limited bench power supply	3.6 V stand-in for the D cells	1	–	Lab (borrow)
Oscilloscope	Inrush, rail dips, preamp signal	1	–	Lab (borrow)
Power profiler (e.g. Nordic PPK2)	Logs µA sleep current over a whole wake cycle (optional, strongly recommended)	1	$0.00–100.00	Lab (borrow)
Reference thermometer	Checks DS18B20 ±0.5 °C	1	$10.00–20.00	Lab (borrow)
Micro-USB + USB-C data cables	Programming the M0 and the ESP32-S3	2	$0.00–20.00	Team (own)
Laptop with Arduino IDE / PlatformIO	Firmware upload, serial monitor	1	–	Team (own)
Subtotal — Test and measurement			**$0 · fallback $10.00–140.00**	
C · Prototyping				
Full-size breadboards	Bench builds, buoy and shore	2	$0.00–20.00	Team (own)
Jumper wire kit + alligator clips	Bench wiring and probing	1	$0.00–10.00	Team (own)
Subtotal — Prototyping			**$0 · fallback $0.00–30.00**	
D · Water testing				
Test tank or bucket (≈ 20 L)	Sensor tests in water	1	–	Lab (borrow)
Sea salt + hydrometer	Seawater at the right salinity	1	$10.00–18.00	Lab (borrow)
Kaolin clay (or clean soil)	Clear vs turbid water check	1	$0.00–8.00	Lab (borrow)
Ice + warm water	Temperature step tests	1	–	Team (own)
Subtotal — Water testing			**$0 · fallback $10.00–26.00**	

ITEM	WHAT IT'S FOR	QTY	COST IF NOT IN LAB	SOURCE
E · Safety				
Safety glasses	Soldering, lithium cells	1	$0.00–8.00	Lab (borrow)
Fire-resistant battery bag	Storing the lithium D cells	1	$10.00–15.00	Lab (borrow)
ESD wrist strap + mat	Protecting the boards	1	$0.00–20.00	Lab (borrow)
Subtotal — Safety			**$0 · fallback $10.00–43.00**	
TOTAL expected equipment spend (all borrowed from the lab)			**$0.00**	
Fallback only, if the lab is missing an item			**$59.00–324.00**	

Sheet 6 — General notes, checks and references

General notes

1. All logic and sensors run at 3.3 V from the Feather's AP2112K regulator. No 5 V rail in the default build (MiniBoost 4654 is a backup only).
2. Remove the Feather's MCP73831 charger before any primary lithium cell is connected. Bench testing uses a current-limited 3.6 V supply instead of the cells.
3. Batteries connect in parallel through one ideal diode each, then the PTC fuse, then the keyed JST-PH plug.
4. Turbidity board and hydrophone circuit are powered only while sampling (Q1 on D11, Q2 on A3, LOW = on; 100 kΩ pull-ups keep them off during sleep and boot).
5. Hydrophone bias: 3.3 V through R9 100 Ω + R4 1.1 kΩ gives ≈ 2.46 V at the plug at 0.7 mA (minimum 2 V), matching Aquarian's test condition (2.5 V compliance, 1.1 kΩ).
6. Expected hydrophone level: −172 dB re 1 V/µPa at a 120 dB re 1 µPa reef soundscape gives ≈ 2.5 mV; preamp gain ≈ 230 brings it to ≈ 0.6 V for the 3.3 V ADC †.
7. Audio is stored on the SD card only; LoRa carries summary numbers (one 1-minute clip would need hours of airtime).
8. Never transmit without an antenna attached. Deployed antenna gain must not exceed 2.15 dBi (RFM95 FCC grant).
9. I²C addresses are unique (0x44 SHT40, 0x68 PCF8523). SPI chip selects: D8 radio, D10 SD; only one LOW at a time.
10. Firmware polarity: config.h says the sensor gate is HIGH = on, but the P-FET switches here are LOW = on. Firmware must drive D11 and A3 LOW to power the sensors.
11. DS18B20 data stays on D12 (firmware). The Rev A schematic label D5 is outdated; D5 stays the PCF8523 interrupt in firmware. D6 is left free in case the radio IO1 pad is ever jumpered for LoRaWAN.

Checked against the sources

✓ H2dM: −172 dB, 1.1 kΩ output, 0.7 mA, min PIP 2 V, 3.5 mm TRS dual-mono (Aquarian H2dM manual p. 1–2)

✓ H2dM price $229 base; 3 / 6 / 9 / 15 m cables (aquarianaudio.com H2d page)

✓ DS18B20: 3.0–5.5 V, 4.7 kΩ pull-up, 750 ms (Maxim datasheet p. 6, 8, 19)

✓ Grove turbidity: 3.3 V / 5 V, output falls as turbidity rises (Seeed wiki)

✓ SHT40: I²C 0x44, 10 kΩ pull-ups on board (Adafruit guide)

✓ Adalogger: RTC on I²C, SD CS = GPIO 10 on M0 (Adafruit guide p. 6–7)

✓ Feather M0: BAT divider 100 k/100 k to D9 = AIN7; radio D3/D4/D8 (schematic rev B)

✓ DMG2305UX: P-channel, −20 V, $R_{DS(on)}$ ≤ 100 mΩ at −2.5 V, $V_{GS(th)}$ −0.5 to −0.9 V, V_{GS} ±8 V max (Diodes DS36196 p. 1–2, saved in DATASHEETS)

✓ Adalogger schematic: SD CS on GPIO10 with 100 k pull-up, 10 k I²C pull-ups, RTC INT1 on an unconnected pad (DATASHEETS/feather_schem.png)

✓ Adafruit 851 = uFL to panel-mount SMA (not RP-SMA), 15 cm RG178; 1661 = uFL SMT connector, $0.75 (adafruit.com)

✓ A1–A5 work as digital I/O; radio uses #8/#3/#4 internally (Adafruit Feather M0 LoRa pinout page)

✓ Adafruit prices from invoice 3758294 and adafruit.com (2026-10-06)

Not yet verified (marked †)

Preamp resistor and capacitor values, PTC rating, ideal-diode part, stack height, Grove current draw, Saft cell price at the chosen seller, whip antenna part number.

References

Aquarian Audio H2d hydrophone page and H2dM user's guide (saved in DATASHEETS) · Maxim DS18B20 datasheet rev 042208 · Seeed Grove turbidity wiki · Adafruit SHT40 guide · Adafruit Adalogger FeatherWing guide · Adafruit Feather M0 RFMxx schematic rev B · Semtech SX1276 datasheet · S.C.O.U.T. Power Budget Update 2026-10-05 and Electronics Build and Test Plan.

Sheet 7 — Every connection: parts, values, wires and how to join them

One table per circuit. Every resistor, capacitor, wire and joint on the build is listed here with its reference (R1, C1 …), value, both ends, and how it is joined. Build each circuit on a breadboard first. Solder it onto the FeatherWing Proto only after it passes its stage in the Electronics Build and Test Plan.

Joining methods: **Solder** means soldered on the Proto board. **Terminal** means the screw terminal (no solder on the probe wire). **Plug** means a connector. **HS** means slide heat-shrink over before soldering, then shrink it.

7A · DS18B20 temperature probe → D12 (Stage 1)

REF	PART / VALUE	FROM	TO	HOW
W1-red	Probe VDD wire	Probe	T1 pin 1 → 3V rail	Terminal
W1-yel	Probe data wire (yellow or white, depending on batch)	Probe	T1 pin 2 → **D12**	Terminal
W1-blk	Probe GND wire (black or blue, depending on batch)	Probe	T1 pin 3 → GND rail	Terminal
R1	4.7 kΩ (comes in the probe bag)	T1 pin 2 (data)	3V rail	Solder
T1	3-pin screw terminal, 3.5 or 5 mm pitch	Proto board edge		Solder

Prep: thread the cable through gland G1 first. Strip 5 mm off each wire, twist the strands, and tin them only lightly so the screw still bites. Check with a meter: about 3.3 V between red and black, and about 3.3 V on data at idle because of the pull-up.

7B · Turbidity sensor and its power switch Q1 → A1, D11 (Stage 1 direct, Stage 3 switched)

REF	PART / VALUE	FROM	TO	HOW
Q1	DMG2305UX P-MOSFET on SOT-23 adapter	S = 3V rail · G = **D11**	D = TURB_3V3 node	Solder (MOSFET to adapter first)
R2	100 kΩ	Q1 gate	3V rail (keeps it OFF)	Solder
C1	100 µF 10 V electrolytic	TURB_3V3 (+ lead)	GND (– stripe)	Solder, watch polarity
C2	0.1 µF ceramic	TURB_3V3	GND, next to Q1	Solder
W3-red	Grove cable VCC	Grove board	TURB_3V3	Solder pin to Proto pad
W3-blk	Grove cable GND	Grove board	GND rail	Solder

REF	PART / VALUE	FROM	TO	HOW
W3-yel	Grove cable SIG	Grove board	**A1** (no divider)	Solder
W3-wht	Grove cable NC	Not used: cut it short and insulate the end		HS

Set the switch on the Grove board to **A** (analog). Firmware: D11 LOW = on (SCO-186). At turn-on, check the 3V rail on the scope. If it dips more than about 100 mV, change C1 to 10 µF †.

7C · SHT40 humidity / leak sensor → I²C (Stage 1)

REF	PART	WIRE COLOUR → PROTO PAD	HOW
W5	STEMMA QT → male cable (Adafruit 4209)	red → 3V · black → GND · blue → SDA · yellow → SCL	Plug on the sensor end, solder the pins on the Proto end

No extra parts needed: the SHT40 and the Adalogger already have their I²C pull-ups. Mount the SHT40 high in the housing, away from the batteries.

7D · Hydrophone front-end → A2, A3 (Stage 7)

REF	PART / VALUE	FROM	TO	PURPOSE
Q2	DMG2305UX on SOT-23 adapter	S = 3V rail · G = **A3**	D = AUD_3V3	Audio power only while recording
R3	100 kΩ	Q2 gate	3V rail	Keeps it OFF while asleep
C8, C9	0.1 µF ceramic + 10 µF	AUD_3V3	GND	Rail decoupling, C8 at U1 pin 8
R9	100 Ω	AUD_3V3	BIAS_F node	Filters rail noise out of the bias
C10	10 µF	BIAS_F	GND	Filters rail noise out of the bias
R4	1.1 kΩ	BIAS_F	J1 **L** (= tip)	Plug-in power: tip sits at ≈ 2.46 V, above the 2 V minimum
J1	3.5 mm jack breakout (Adafruit 1699)	L pin = tip (signal) · GND pin = sleeve	R pin (ring): leave unconnected	Hydrophone plugs in here
C3	1 µF ceramic (non-polar)	J1 L (tip)	U1 pin 3 (IN A+)	Blocks the 2.46 V DC, passes the sound
R5, R6	10 kΩ + 10 kΩ	AUD_3V3 → VMID	VMID → GND	Makes 1.65 V mid-rail
C4	10 µF	VMID	GND	Keeps the mid-rail quiet
R7	100 kΩ	VMID	U1 pin 3	Biases stage 1 at 1.65 V
R10	20 kΩ	U1 pin 1 (OUT A)	U1 pin 2 (IN A–)	Stage 1 gain = 1 + 20k/1k = 21

REF	PART / VALUE	FROM	TO	PURPOSE
R11 + C5	1 kΩ in series with 10 µF	U1 pin 2	GND	Sets the AC gain; DC gain stays 1. Low cut ≈ 16 Hz
R8	1.5 kΩ	U1 pin 1	U1 pin 5 (IN B+)	Low-pass filter ≈ 10.6 kHz (anti-alias)
C6	10 nF ceramic	U1 pin 5	GND	Low-pass filter ≈ 10.6 kHz (anti-alias)
R12	10 kΩ	U1 pin 7 (OUT B)	U1 pin 6 (IN B–)	Stage 2 gain = 1 + 10k/1k = 11
R13 + C7	1 kΩ in series with 10 µF	U1 pin 6	GND	Stage 2 AC gain; DC stays at 1.65 V
U1	MCP6002 DIP-8 in a socket	pin 8 → AUD_3V3 · pin 4 → GND	pin 7 → **A2**	Total gain ≈ 231 (47 dB)

The electrolytic 10 µF caps C4, C5 and C7 sit at 1.65 V, so their + lead goes toward the op-amp/VMID side and the – stripe to GND. 10 µF ceramics avoid the polarity question. Bench check: with the hydrophone in a bucket and tapped, A2 should swing around 1.65 V without clipping at 0 V or 3.3 V. If it clips, lower R10 to 10 kΩ †.

7E · Battery pack ⟶ Feather BAT (Stage 6; remove the Feather's charger first)

REF	PART / VALUE	FROM	TO	HOW
BT1, BT2	Saft LS33600 with solder tabs	+ tab	D1 / D2 VIN	22 AWG stranded red, solder to the **tab only**, HS
BT1, BT2	Cell – tabs	– tab	Pack GND bus	22 AWG stranded black, HS
D1, D2	LM66100 ideal diode on SC-70-6 adapter †	VIN ← cell + · GND → pack – · CE → GND (enabled, check datasheet) †	VOUT → common + node	Solder
F1	PTC resettable fuse, 0.5 A hold (e.g. Bourns MF-R050) †	Common + node	P1 red	Solder inline, HS
P1	JST-PH 2-pin cable with tinned ends (Adafruit 261)	red ← F1 · black ← pack –	Plugs into the Feather's battery jack	Solder + HS, then plug

Never solder to the cell body and never short a cell. Before plugging in, measure P1: red should read +3.5 to 3.7 V relative to black. Cable-tie and foam-pad both cells in the bay, and store spares in the fire bag.

7F · Antenna and board stack (Stages 1, 4, 8)

REF	PART	WHAT TO DO
H1, H2	Stacking headers 2830 (×2)	Solder onto the M0 and the Adalogger, sockets facing up. Keep the pins straight by plugging them into a breadboard while soldering.
H3	Male headers, 12 + 16 pin	Solder on the bottom of the Proto board so it plugs into the Adalogger sockets. Use the board's own headers or lab stock.
CR1	CR1220 coin cell	Into the Adalogger holder, + side up.
ANT-b	Spring antenna 4269 (bench)	Solder to the M0 ANT pad. Remove it before fitting the uFL path.
ANT-d	uFL 1661 → cable 851 → whip	Solder 1661 to the M0 uFL footprint (fine tip, flux). Snap on 851, put its SMA through the lid and tighten the nut, then screw on the whip.
SS	Shore: ESP32-S3 + LoRa wing	Stacking headers on the ESP32-S3. On the wing, bridge the CS, RST and IRQ jumpers to the pins in the firmware, and solder the spring antenna.

Build order (follows the Electronics Build and Test Plan)

1. **Stage 1:** headers on the M0 and the Adalogger, coin cell in. On a breadboard: DS18B20 + R1, SHT40 via 4209, turbidity straight to 3V and A1. Get first readings over USB.
2. **Stage 2:** measure sleep and active current with the meter in series.
3. **Stage 3:** build Q1 + R2 + C1 + C2 on the breadboard and check that D11 LOW turns the sensor on. Then remove the Feather's charger.
4. **Stage 4:** spring antennas on both radios; shore jumpers bridged. Range test.
5. **Stage 6:** build the battery pack (7E), check its polarity, then the 7-day run.
6. **Stage 7:** breadboard the hydrophone front-end (7D), check A2, record test clips.
7. **Stage 8:** move all circuits onto the Proto board, terminals and jack at the edges. Thread cables through the glands before terminating. Fit the uFL → SMA path. Conformal-coat the boards (keep coating off the connectors and the SD slot), add desiccant, seal.